```python
# Naive Bayes Classification for Bank Loan Prediction

# Import libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Create sample bank loan dataset
data = {
    'Income': [25000, 30000, 35000, 40000, 45000,
               50000, 55000, 60000, 65000, 70000,
               28000, 38000, 48000, 58000, 68000],

    'Loan_Amount': [100000, 120000, 150000, 180000, 200000,
                    150000, 180000, 200000, 220000, 250000,
                    130000, 160000, 180000, 210000, 230000],

    'Credit_Score': [550, 580, 600, 620, 650,
                     680, 700, 720, 750, 780,
                     560, 610, 660, 710, 760],

    'Loan_Approved': [
        0, 0, 0, 0, 1,
        1, 1, 1, 1, 1,
        0, 0, 1, 1, 1
    ]
}
```

```
# Convert to DataFrame
df = pd.DataFrame(data)

# Display dataset
print("Bank Loan Dataset:")
print(df)

# Input features
X = df[['Income', 'Loan_Amount', 'Credit_Score']]

# Target variable
y = df['Loan_Approved']

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.3,
    random_state=42,
    stratify=y
)

# Create Naive Bayes model
model = GaussianNB()

# Train model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)
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# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

print("\nNaive Bayes Bank Loan Prediction")
print("--------------------------------")
print("Accuracy:", accuracy)
print("Accuracy (%):", accuracy * 100)

# Display actual vs predicted
print("\nActual vs Predicted:")

for actual, predicted in zip(y_test, y_pred):
    actual_name = "Approved" if actual == 1 else "Not Approved"
    predicted_name = "Approved" if predicted == 1 else "Not Approved"

    print(actual_name, "->", predicted_name)

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)

# Display confusion matrix
plt.figure(figsize=(6, 4))

sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
```

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    xticklabels=["Not Approved", "Approved"],
    yticklabels=["Not Approved", "Approved"]
)

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Naive Bayes - Bank Loan Prediction")
plt.show()

# Classification Report
print("\nClassification Report:")

print(
    classification_report(
        y_test,
        y_pred,
        target_names=["Not Approved", "Approved"],
        zero_division=0
    )
)
```

```
Bank Loan Dataset:
    Income  Loan_Amount  Credit_Score  Loan_Approved
0    25000       100000           550              0
1    30000       120000           580              0
2    35000       150000           600              0
3    40000       180000           620              0
4    45000       200000           650              1
5    50000       150000           680              1
6    55000       180000           700              1
7    60000       200000           720              1
8    65000       220000           750              1
9    70000       250000           780              1
10   28000       130000           560              0
11   38000       160000           610              0
12   48000       180000           660              1
13   58000       210000           710              1
14   68000       230000           760              1

Naive Bayes Bank Loan Prediction
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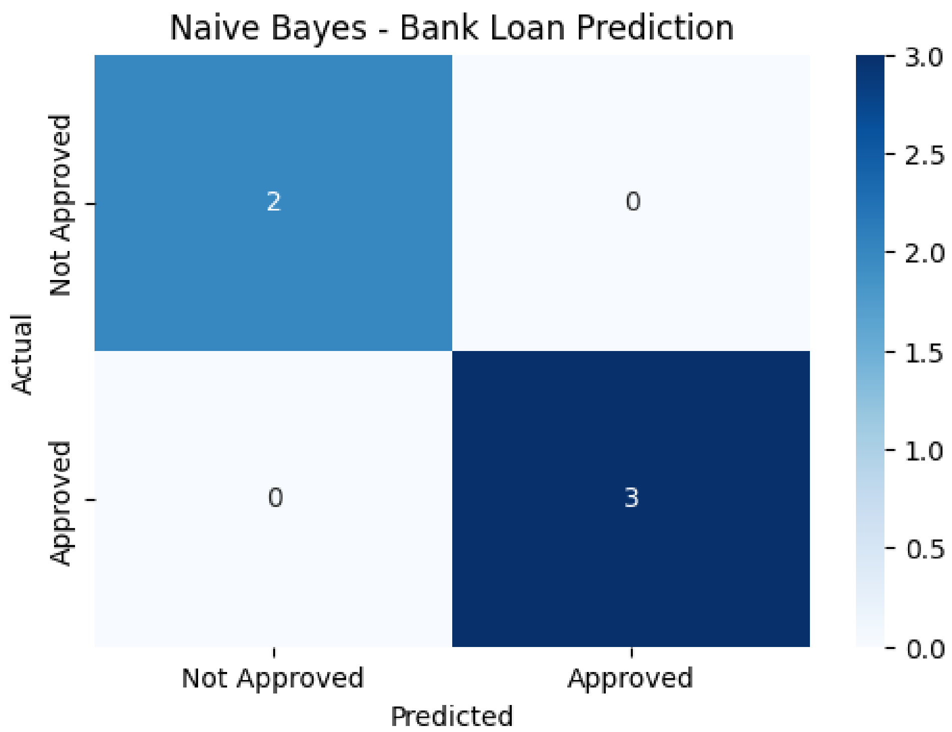
Accuracy: 1.0
Accuracy (%): 100.0
```



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Classification Report:
              precision    recall  f1-score   support

Not Approved       1.00      1.00      1.00         2
    Approved       1.00      1.00      1.00         3

    accuracy                           1.00         5
   macro avg       1.00      1.00      1.00         5
weighted avg       1.00      1.00      1.00         5
```